TEXORA · LECTURE SHEET

Work Measurement: Rating, Basic Time and Standard Time

How long a job should take — the techniques of work measurement, the element breakdown and break points of a time study, performance rating, and the build-up from observed time through basic time to standard time with allowances.

TE-4107 · PRODUCTION PLANNING AND CONTROL · TOPIC 3 · LEVEL 4 TERM 1

1. Purpose and techniques

Work measurement determines the time a qualified worker, using the standard method, needs to complete a job at standard performance. Its purposes: provide standard times for costing and pricing, for production planning and line balancing, for manpower calculation, for incentive schemes, and for comparing alternative methods objectively. Without measured times, every one of those activities runs on opinion.

TECHNIQUE	HOW IT WORKS	BEST SUITED TO
Time study	Stopwatch (or electronic) timing of elements over repeated cycles, with rating	Short-cycle repetitive work — sewing operations, winding, inspection
Predetermined motion time systems (PMTS)	Times built from tabulated standard data for basic human motions (e.g. MTM, GSD in garments)	New styles before production starts; method design
Standard data / synthesis	Times assembled from a bank of previously studied elements	Families of similar jobs — garment operations libraries
Activity sampling	Random instantaneous observations classify activity proportions (Topic 4)	Long-cycle and irregular work; utilisation and delay studies
Analytical estimating	Skilled estimator builds time from experience + measured parts	One-off, non-repetitive work — maintenance, machine setting

2. Time study: elements and break points

A job is divided into **elements** — distinct, separately timed parts with clear start and finish. Elements are chosen so that: machine time and manual time separate; constant and variable parts separate; high- and low-effort parts separate (rating differs between them); and each element is long enough to time reliably (about 0.04 min minimum with a stopwatch). The boundary between elements is the **break point** — an unmistakable event (a sound, a motion, a machine signal) the observer can catch identically every cycle. Poor break points — a vague "somewhere in the reach" — make a study irreproducible: two observers get two different times for the same work.

The observer records the time at each break point over many cycles, discarding cycles disturbed by foreign elements (thread break from a faulty cone, a supervisor's question) — recorded as such, never silently deleted. The average of clean readings is the **observed time** for the element.

3. Rating

Observed time belongs to the operator actually seen, working at that operator's pace. **Rating** converts it to the pace of the *standard* operator: the observer compares the worker's effective speed of movement — not output, but the rate of working — with the standard rating of 100. A worker judged 10 % faster than standard is rated 110; slower, 90. Rating is a trained judgment, calibrated against standard references (walking at 6.4 km/h is the classic 100-rating pace for level walking; dealing 52 cards into four piles in 30 s is another standard reference). Honest rating is the ethical core of time study: inflate the rating and the standard becomes unfairly tight; deflate it and costing absorbs slack forever.

$$\text{Basic time} = \text{Observed time} \times (\text{Rating} \div 100)$$

Basic time (also called normal time) is thus the time the standard operator would take for the element, working at standard pace with no allowances yet.

4. Allowances and standard time

No one works a full shift at uninterrupted standard pace. **Allowances** are added to basic time: **relaxation allowance** for personal needs and recovery from fatigue (varies with posture, load, heat, monotony — a hot, standing, heavy job earns more than a seated light one); **contingency allowance** for small, legitimate irregular work and delays within the job's scope (typically a few per cent); and, where applicable, interference or quality-related allowances. Allowances are expressed as a percentage of basic time (or of shift time — the basis must be stated, as the arithmetic differs):

$$\text{Standard time} = \text{Basic time} \times (1 + \text{Total allowance \%})$$

WORKED EXAMPLE — TIME STUDY OF A SEWING OPERATION

Operation: attach pocket to shirt front (standard method installed, Topic 2). Element study, average observed times and ratings:

ELEMENT	OBSERVED TIME (MIN)	RATING	BASIC TIME (MIN)
1. Pick up panel and pocket, position	0.12	100	0.120
2. Sew pocket, three sides	0.35	110	0.385
3. Trim threads, inspect	0.22	95	0.209
4. Dispose to bundle	0.15	105	0.1575
Total			**0.8715**

Allowances: relaxation 12 % + contingency 3 % = 15 % of basic time.
Standard time = 0.8715 × 1.15 = **1.0022 ≈ 1.00 min** (the SMV of the operation).
Expected output at standard performance = 480 ÷ 1.0022 ≈ **479 pieces per 8-hour shift** per operator.
Check the element ratings as the observer's logic: the machine-paced sewing element was performed briskly (110), the manual trim-and-inspect element more deliberately (95) — different elements, different effective paces, which is exactly why the job is studied element by element rather than as one lump.

5. Uses and limits

Standard times drive: garment costing (CM from SMV), line balancing (Topic 8 of AM-4202 uses the same arithmetic), capacity and manpower planning, delivery promising, and incentive design. Their limits are equally real: they describe the *standard method* only, decay when methods, machines or materials change, and become a source of conflict if used punitively. A standard time is a planning instrument, not a whip — mills that treat it as a whip find their operators managing the stopwatch instead of the work.

Glossary

Work measurement — techniques for establishing the time for a qualified worker at standard performance to carry out a standard method. **Element** — a separately timed part of a job with clear break points. **Break point** — the instant an element ends and the next begins. **Observed time** — average clean time recorded for an element. **Rating** — assessment of the worker's pace against standard performance (100). **Basic (normal) time** — observed time corrected by rating. **Allowance** — addition

to basic time for relaxation, contingencies and related needs. **Standard time / SMV** — basic time plus allowances; the planning time for the job.

Key points

• Basic time = observed time × rating/100; standard time = basic time × (1 + allowances).
• Break points must be sharp events; vague break points make studies irreproducible.
• Worked study: element basic times sum to 0.8715 min; with 15 % allowances SMV ≈ 1.00 min → ≈ 479 pieces/shift.
• Rating judges pace of movement, not output, and is the ethical core of time study.
• Standard times feed costing, balancing, planning and incentives — and expire when the method changes.

Practice

1. An element's observed times over 8 cycles average 0.46 min at rating 90. Find the basic time. If total allowances are 18 %, find the standard time and the standard output per 480-min shift.
2. Why are machine-controlled elements rated differently in principle from manual elements?
3. A supervisor complains that two observers produced different standard times for the same job. Give three likely technical causes.
4. State the allowance categories in the worked example and explain why a standing job in a hot dye house should carry a larger relaxation allowance than a seated inspection job.

Answer key

1. Basic time = 0.46 × 90/100 = 0.414 min. Standard time = 0.414 × 1.18 = 0.4885 ≈ 0.49 min. Output = 480 ÷ 0.4885 ≈ 983 units per shift.
2. A machine-controlled element takes the time the machine takes, whoever tends it — the operator cannot hurry a loom's picks or a dyeing machine's programmed ramp. Rating such an element at other than 100 distorts the study: the machine portion must enter basic time at its actual (100-rated) duration, while only the manual portions carry the operator's rating. Mixing them hides where time is truly controllable.
3. (i) Different element breakdowns or vague break points, so the observers timed different content; (ii) rating differences — untrained or uncalibrated observers judging pace differently; (iii) different treatment of foreign elements and disturbed cycles (one included them, the other excluded them), or timing during different method versions — the method itself had drifted between studies. (Also acceptable: too few cycles observed, so averages are unstable.)

4. Relaxation allowance (12 %) and contingency allowance (3 %). Relaxation allowance repays the energy and recovery a job demands: standing posture, heat load from steam and hot liquors, and heavier physical effort all raise fatigue and the rest needed to sustain standard performance across a shift. A seated inspector in a conditioned room has low energy expenditure and full postural relief, so the same percentage would overstate need — allowances are set by job conditions, not by habit.

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